# Dynamic Partial Cointegration (Dynamic-PCI) & Neural Kalman Filtering for Statistical Arbitrage

Comprehensive Research Report, System Architecture, Empirical Results, and Execution Audit

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**Abstract**

This report provides a rigorous, end-to-end engineering and quantitative audit of the **Dynamic Partial Cointegration (Dynamic-PCI)** algorithmic pairs trading framework. The system resolves two pervasive failure modes in classical statistical arbitrage: time-varying cointegrating hedge ratios $\beta_t$ and the contamination of mean-reverting price spreads by permanent, non-stationary structural drift $R_t$. We detail the mathematical formulations of both the 4-state joint state-space model and the reduced 3-state model with exogenous driver prices, examine the integration of recurrent neural Kalman gains via **KalmanNet** trained under a **Differentiable Sharpe Ratio Loss**, present the online dual-gating risk-control engine, and analyze empirical out-of-sample backtest results across a 14-pair cross-asset universe (11 core benchmark pairs plus 3 newly validated passing pairs) spanning 12 years (2009–2021), supported by an extended 21-pair screening research study. Full root-cause analyses of prior failure modes (cascaded filtering, transition matrix bugs, and BPTT gradient explosion / NaN losses) are documented alongside formal academic citations and future research trajectories.

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# 1 Executive Summary & Project Evolution

Pairs trading rests on identifying economically or statistically linked assets whose spread temporarily widens, anticipating reversion to equilibrium. Real-world financial time series violate classical Engle–Granger cointegration assumptions in two fundamental ways:

1. **Dynamic Hedge Ratios:** The true equilibrium coefficient $\beta_t$ drifts due to evolving firm balance sheets, structural commodity shocks, and market regime transitions. Static OLS or static Engle–Granger regressions fail over multi-year horizons.
2. **Permanent Drift vs. Transient Dislocations:** Real-world price spreads consist of both a **transient mean-reverting component** ($M_t$) and a **permanent non-stationary random-walk drift** ($R_t$). Strategies treating raw spread widening as a discount face catastrophic drawdowns during structural breaks.

To solve both challenges simultaneously, this project underwent four distinct evolutionary phases:

- **v1 Generation (Decoupled & Naive 4-State Formulations):** Attempted two-stage cascaded filtering (estimating $\beta_t$ via one filter and decomposing residuals via a second), followed by an unconstrained naive 4-state joint filter $[X_{1,t}, \beta_t, M_t, R_t]^T$. Both suffered from implementation defects, including an inverted transition matrix $\mathbf{F}$, static observation equations, and gradient explosion during backpropagation through time.
- **Audit & Stabilization Phase:** Audited critical mathematical and architectural defects, established hard state clamping to eliminate numerical overflow and NaN loss during neural training, and introduced scale-free Base-1.0 price normalization.
- **v2 Experimental Baseline (`pipeline_v2.py`):** Explored the reduced 3-state state-space formulation with exogenous driver pricing, an online Dual-Gate, and regularized KalmanNet execution (preserved in `archive/`).
- **Consolidated Production Framework (`main.py`):** Standardized on the stabilized state-space model with dynamic observation matrix (`dynamic_pci.py`), an automated 4-criterion pre-trade screening engine (`regime_filter.py`), two-stage Neural KalmanBOT training with Differentiable Sharpe Loss (`neural_kalman_pci.py`), and a unified master orchestrator providing complete dual-logging and benchmarking across the complete 14-pair cross-asset universe.

# 2 Theoretical Foundations & Economic Assumptions

## 2.1 Driver Asymmetry & Scale Separation

Inverting the asset order $X_1 \leftrightarrow X_2$ induces non-linear behavior ($\beta' \approx 1/\beta$). Designating a volatile, illiquid follower as $X_1$ destabilizes $\beta_t$. Hence, the larger, higher-volume asset is consistently fixed as $X_1$. Furthermore, the framework enforces **scale separation**: the hedge ratio process noise $q_\beta$ is constrained so that $\beta_t$ adapts only to low-frequency macroeconomic trends, preventing it from absorbing high-frequency deviations belonging to $M_t$.

# 3 Mathematical Models & State-Space Architectures

## 3.1 State-Space Formulations: 4-State Joint System & Reduced 3-State Model

The framework admits two mathematically aligned state-space representations:

Table 1: Theoretical Comparison: Classical Cointegration vs. Proposed Dynamic-PCI

| Dimension | Classical Cointegration (Engle–Granger) | Dynamic-PCI Framework (Our System) |
|---|---|---|
| **Hedge Ratio ($\beta$)** | Static constant $\beta \in \mathbb{R}$ estimated via OLS. | Dynamic random walk $\beta_t = \beta_{t-1} + w_t^{(\beta)}$ with step variance $q_\beta^2$. |
| **Spread Dynamics** | Assumes raw spread $y_t - \beta x_t \sim I(0)$ is strictly stationary. | Spread decomposes: $y_t - \beta_t x_t = M_t + R_t$, where $M_t \sim \text{AR}(1)$ and $R_t \sim I(1)$. |
| **Driver Role** | Arbitrary assignment of dependent vs. independent leg. | High-liquidity sector leader assigned as exogenous driver $X_1$; follower as response $X_2$. |
| **Observation Model** | Direct linear combination without noise. | Response $X_2$ observed with measurement noise $v_t \sim \mathcal{N}(0, \sigma_{\text{obs}}^2)$. |
| **Market Frictions** | Zero transaction costs assumed (0 bps). | Exact 10 bps (0.0010) per half-turn applied to both legs on every portfolio rebalance. |

### 3.1.1 The Production 4-State Joint Model (`dynamic_pci.py`)

In the production implementation, the state vector couples the driver asset price, the dynamic cointegrating hedge ratio, the mean-reverting spread, and the permanent random-walk drift:

$$\mathbf{z}_t = \begin{bmatrix} X_{1,t} \\ \beta_t \\ M_t \\ R_t \end{bmatrix} \in \mathbb{R}^4, \qquad \mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \rho & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{Q} = \text{diag}(\sigma_{X_1}^2, q_\beta^2, \sigma_M^2, \sigma_R^2) \tag{1}$$

The observation equation couples response $X_{2,t}$ and driver $X_{1,t}$ via a dynamic 2D measurement system:

$$\mathbf{y}_t = \begin{bmatrix} X_{2,t} \\ X_{1,t} \end{bmatrix} = \mathbf{H}_t \mathbf{z}_t + \mathbf{v}_t, \qquad \mathbf{H}_t = \begin{bmatrix} 0 & X_{1,t} & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \qquad \mathbf{R}_{\text{obs}} = \text{diag}(\sigma_{\text{obs},2}^2, \sigma_{\text{obs},1}^2) \tag{2}$$

where the second row enforces the exact driver identity $X_{1,t} = X_{1,t} + v_{1,t}$. When the driver measurement noise $\sigma_{\text{obs},1}^2 \to 0$, $X_{1,t}$ is observed without uncertainty, collapsing the 4-state system into an equivalent reduced 3-state model.

### 3.1.2 The Equivalent Reduced 3-State Model (Exogenous Driver)

Conditioning directly on observed driver price $X_{1,t}$ as an exogenous time-varying regressor reduces the latent state to $\mathbf{z}_t = [\beta_t, M_t, R_t]^T \in \mathbb{R}^3$:

$$X_{2,t} = \mathbf{H}_t \mathbf{z}_t + v_t, \qquad \mathbf{H}_t = \begin{bmatrix} X_{1,t} & 1 & 1 \end{bmatrix}, \qquad v_t \sim \mathcal{N}(0, \sigma_{\text{obs}}^2) \tag{3}$$

with transition matrix $\mathbf{F} = \text{diag}(1, \rho, 1)$ and process covariance $\mathbf{Q} = \text{diag}(q_\beta^2, \sigma_M^2, \sigma_R^2)$. Both formulations yield identical spread filtering dynamics under Base-1.0 normalization.

### 3.1.3 Kalman Filter Recursion & Maximum Likelihood Estimation

$$\hat{\mathbf{z}}_{t|t-1} = \mathbf{F}\hat{\mathbf{z}}_{t-1|t-1} \tag{4}$$
$$\mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^T + \mathbf{Q} \tag{5}$$
$$\mathbf{e}_t = \mathbf{y}_t - \mathbf{H}_t\hat{\mathbf{z}}_{t|t-1} \tag{6}$$
$$\mathbf{S}_t = \mathbf{H}_t\mathbf{P}_{t|t-1}\mathbf{H}_t^T + \mathbf{R}_{\text{obs}} \tag{7}$$
$$\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}_t^T\mathbf{S}_t^{-1} \tag{8}$$
$$\hat{\mathbf{z}}_{t|t} = \hat{\mathbf{z}}_{t|t-1} + \mathbf{K}_t\mathbf{e}_t \tag{9}$$
$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{H}_t)\mathbf{P}_{t|t-1} \tag{10}$$

Parameters $\boldsymbol{\theta}_{\text{dyn}} = [\beta_0, \rho, \ln\sigma_M, \ln\sigma_R, \ln q_\beta]$ are estimated on formation data via prediction-error decomposition Maximum Likelihood Estimation (MLE) using L-BFGS-B (with observation noise fixed analytically to $\sigma^2_{\text{obs}} = (0.10 \times \sigma_{X_1})^2$ to guarantee well-conditioned gain updates):

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2}\sum_{t=1}^{T_{\text{form}}}\left[\ln|2\pi\mathbf{S}_t| + \mathbf{e}_t^T\mathbf{S}_t^{-1}\mathbf{e}_t\right] \tag{11}$$

## 3.2 KalmanNet Recurrent Neural Architecture (`neural_kalman_pci.py`)

To adapt to fat-tailed financial distributions where Gaussian noise assumptions fail, KalmanNet replaces analytical gain computation $\mathbf{K}_t$ with a recurrent neural network:

- **Input Feature Vector:** In the canonical KalmanNet framework (Revach et al., 2022), the network input feature vector $\mathbf{u}_t$ is composed of: (1) the normalized innovation vector $\tilde{\mathbf{e}}_t$, and (2) the forward state update from the previous time step $\Delta\hat{\mathbf{z}}_{t-1} = \hat{\mathbf{z}}_{t-1|t-1} - \hat{\mathbf{z}}_{t-1|t-2} = \mathbf{K}_{t-1}\mathbf{e}_{t-1}$ (termed feature $F_4$). *Note on State Deltas:* The difference between the prior state prediction and its propagation $\hat{\mathbf{z}}_{t|t-1} - \mathbf{F}\hat{\mathbf{z}}_{t-1|t-1}$ is identically zero by definition of the state transition equation ($\hat{\mathbf{z}}_{t|t-1} \equiv \mathbf{F}\hat{\mathbf{z}}_{t-1|t-1}$); hence KalmanNet's $F_4$ feature strictly uses the *previous step's update* (posterior minus prior). *Note on Scalar Observation Normalization:* When observation $y_t$ is scalar ($n = 1$), naive L2 normalization $\frac{e_t}{|e_t|+\epsilon}$ reduces directly to $\text{sign}(e_t)$, discarding the vital pricing error magnitude. To retain scale awareness, the production network utilizes the 2D coupled observation vector $\mathbf{y}_t = [X_{2,t}, X_{1,t}]^T \in \mathbb{R}^2$ (preserving relative vector geometry) and incorporates the prior state vector.
- **Input Feedforward Layer:** Linear projection from $(m + n)$ to $H_1 = \min(48, (m + n) \times 8)$ with ReLU activation.
- **Recurrent Core (GRU):** Single-layer Gated Recurrent Unit (GRU) with input_size $= H_1$ (32 in the reduced 3-state exogenous model, or 48 in the 4-state joint system) and hidden_dim $= 32$. This totals exactly $3 \times (32 \times 32 + 32 \times 32 + 2 \times 32) = 6{,}336$ trainable parameters for $H_1 = 32$ (7,872 parameters for $H_1 = 48$) across its reset, update, and candidate state gates.
- **Output Head:** Hidden projection $H_2 = \min(24, \max(8, 2mn))$ followed by linear mapping to $\mathbb{R}^{m\times n}$. Output layer weights and biases are initialized to zero ($\mathbf{W}_{\text{out}} = \mathbf{0}, \mathbf{b}_{\text{out}} = \mathbf{0}$), ensuring initial filter predictions default conservatively to prior state estimates.

# 4 Training Protocol & Loss Formulations

KalmanNet undergoes a two-stage curriculum:

### 4.1 Stage 1: Observation MSE Pre-training

Pre-trains network weights for 10–15 epochs to minimize one-step prediction error on response series $X_{2,t}$:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{T}\sum_{t=1}^{T}\left(X_{2,t} - \mathbf{H}_t\hat{\mathbf{z}}_{t|t-1}\right)^2 \tag{12}$$

### 4.2 Stage 2: Differentiable Sharpe Ratio Fine-Tuning

Directly maximizes risk-adjusted return through Backpropagation Through Time (BPTT). To permit gradient backpropagation through discrete thresholding rules, trading positions are modeled via a continuous hyperbolic tangent proxy:

$$\widetilde{\text{pos}}_t = -\tanh\left(\frac{z(M_t)}{\tau_{\text{soft}}}\right) \tag{13}$$

$$R_t^{\text{strat}} = \widetilde{\text{pos}}_{t-1}\cdot r_t^{\text{pair}} - c\cdot|\widetilde{\text{pos}}_t - \widetilde{\text{pos}}_{t-1}| \tag{14}$$

$$\mathcal{L}_{\text{Sharpe}} = -\frac{\mathbb{E}[R_t^{\text{strat}}]}{\sqrt{\text{Var}(R_t^{\text{strat}}) + 10^{-8}}}\times\sqrt{252} \tag{15}$$

where $c = 0.0010$ (10 bps) represents the transaction cost penalty, penalizing excessive portfolio turnover during gradient descent.

### 4.3 Numerical Stabilization Against Exploding Gradients & NaNs

Recurrent BPTT through Kalman state updates creates an unstable feedback loop: unconstrained gain outputs $\mathbf{K}_t$ inflate state updates $\mathbf{x}_{t|t}$, causing subsequent innovations $e_{t+1}$ and the tanh position derivative to explode into numerical overflow ('NaN'). The issue was permanently resolved by:

1. Enforcing hard forward state update clamping: $\hat{\mathbf{z}}_{t|t} \leftarrow \text{clip}(\hat{\mathbf{z}}_{t|t}, -50.0, 50.0)$.
2. Regularizing GRU capacity down from 200 units to hidden_dim $= 32$.
3. Applying gradient norm clipping: $\|\mathbf{g}\|_2 \leq 1.0$.
4. Detaching recurrent hidden state across batch boundaries: $\mathbf{h}_{t-1} \leftarrow \mathbf{h}_{t-1}$.detach().

## 5 Statistical Screening & Online Regime Gating

### 5.1 Production Pre-Trade Screening Engine (`regime_filter.py`)

Before allocating capital to any pair, candidate time series must pass an automated 4-criterion suitability screen evaluated over the formation window:

1. **Mean-Reverting Variance Ratio ($R^2_{\text{MR}} \geq 15\%$):** Formulated strictly as the Clegg & Krauss (2018) variance ratio of differences:

$$R^2_{\text{MR}} = \frac{\text{Var}(\Delta M_t)}{\text{Var}(\Delta M_t) + \text{Var}(\Delta R_t)} = \frac{2\sigma_M^2}{2\sigma_M^2 + (1+\rho)\sigma_R^2} \tag{16}$$

   Because $R_t \sim I(1)$ has no finite unconditional level variance, differences $\Delta M_t$ and $\Delta R_t$ are both stationary $I(0)$, ensuring the ratio is well-defined and scale-invariant.

2. **Finite Mean-Reversion Half-Life ($HL \leq 252.0$ days):** Calculated as $HL = -\frac{\ln 2}{\ln|\rho|}$, ensuring dislocations revert within one operational trading year.

3. **Spread Stationarity Confirmation ($p_{\mathbf{ADF}} \leq 0.10$):** Rejects the unit-root null hypothesis on the formation spread at the 10% significance level.

4. **Non-Boundary AR(1) Persistence ($\rho < 0.999$):** Prevents numerical optimization collapse toward an unconstrained random walk.

*Multi-Tier Relaxation Framework:* As detailed in Section 9.1, the strict baseline ADF requirement on the raw static spread rejects valid partial cointegration relationships due to multi-year $\beta_t$ drift. Section 9.1 formalizes the Relaxed Tier 1 ($R^2_{\text{MR}} \geq 15\%, HL \leq 365\text{d}, p_{\text{ADF}} \leq 0.20$) and the PCI-Aware Tier ($R^2_{\text{MR}} \geq 10\%, HL \leq 365\text{d}, p_{\text{ADF}}(M_t) \leq 0.05$).

### 5.2 Online Execution Gating (`backtest_v2.py`)

- **Deterministic Volatility Gate:** Suspends trading if short-term spread volatility exceeds $3\times$ baseline:
$$gate_{\text{vol},t} = \mathbb{I}\left(\frac{\text{std}(M_{t-20:t})}{\text{std}(M_{1:t})} \leq 3.0\right) \tag{17}$$

- **Rolling AR(1) Regime Gate:** Fits rolling 60-day OLS $\Delta M_t = \kappa(\mu - M_{t-1}) + \epsilon_t$. If $HL_{\text{roll}} > 126$ days or $\kappa \leq 0$, sets $gate_{\text{AR1},t} = 0$.

- **Composite Gate:** $gate_t = gate_{\text{vol},t} \times gate_{\text{AR1},t}$. When $gate_t = 0$, open positions are closed flat and new entries are blocked.

## 6 Trading Strategy & Execution Backtesting

### 6.1 Signal Construction & Execution Rules

Trading signals are generated exclusively from the filtered transient component $M_t$, completely decoupling execution from permanent drift $R_t$:

$$z_t = \frac{M_t - \mu_M(W)}{\sigma_M(W)}, \quad W = 80 \text{ days} \tag{18}$$

- **Long Pair Entry:** $z_t \leq -\tau_{\text{open}}$ and $gate_t = 1 \implies$ Long $X_2$, Short $\beta_t X_1$ (production benchmark $\tau_{\text{open}} = 1.0\sigma$).

- **Short Pair Entry:** $z_t \geq +\tau_{\text{open}}$ and $gate_t = 1 \implies$ Short $X_2$, Long $\beta_t X_1$.

- **Full Equilibrium Exit:** Dislocation reverts across zero ($z_t \geq 0.0$ for long spreads, $z_t \leq 0.0$ for short spreads) $\implies$ Close position flat to cash ($\text{pos}_t = 0$).

- **Catastrophic Stop-Loss:** Position drawdown $\leq -10\%$ triggers immediate emergency liquidation.

- **Comparative Rule Ablations (Section 9.3):** The Clegg & Krauss (2018) early profit-taking rule ($|z_t| \leq 0.5\sigma$, closing 50% or 100% early) and conservative thresholds ($\tau_{\text{open}} = 1.5\sigma$) are analyzed as comparative ablations in Section 9.3.

### 6.2 Dollar-Neutral Sizing & Net Return Accounting

Positions maintain continuous dollar-neutral weights determined dynamically from posterior hedge ratios $\beta_{t-1}$:

$$w_{2,t-1} = \frac{1}{1+|\beta_{t-1}|}, \qquad w_{1,t-1} = \frac{\beta_{t-1}}{1+|\beta_{t-1}|} \tag{19}$$

where net daily portfolio returns account for 10 bps ($c = 0.0010$) transaction friction on each asset leg whenever position exposure changes ($2c = 20$ bps round-trip on rebalances):

$$r_t^{\text{pair}} = \text{pos}_{t-1}\left(w_{2,t-1}\frac{X_{2,t}-X_{2,t-1}}{X_{2,t-1}} - w_{1,t-1}\frac{X_{1,t}-X_{1,t-1}}{X_{1,t-1}}\right) - 2c\cdot\mathbb{I}(\text{pos}_t \neq \text{pos}_{t-1}) \tag{20}$$

with cumulative strategy equity compounding as $E_t = \prod_{s=1}^{t}(1+r_s^{\text{pair}})$.

# 7 Comparative Methodologies: Step-by-Step Architectural Walkthrough

To rigorously evaluate the relative merits of static cointegration, time-varying state-space tracking, regime-gated risk control, and deep recurrent filtering, this project benchmarks four distinct quantitative methodologies across the exact same price datasets. Below is a comprehensive, step-by-step operational and mathematical walkthrough of each framework.

### 7.1 Method 1: Static-PCI (Fixed-Beta Clegg & Krauss Baseline)

The Static Partial Cointegration (Static-PCI) model represents the academic benchmark established by Clegg & Krauss (2018). It decomposes price spreads into mean-reverting and random-walk states while holding the cointegrating vector $\beta_0$ strictly constant.

**Step 1: In-Sample Formation Parameter Estimation.** On the formation period (first 2,014 trading days), parameters $\boldsymbol{\theta}_{\text{static}} = [\beta_0, \rho, \ln\sigma_M, \ln\sigma_R]$ are estimated via Maximum Likelihood Estimation (MLE) using the L-BFGS-B optimizer, fixing the hedge-ratio process noise to zero ($q_\beta^2 \equiv 10^{-12}$).

**Step 2: Prior State & Covariance Propagation.** At each out-of-sample time step $t$, the state vector $\mathbf{z}_t = [\beta_t, M_t, R_t]^T \in \mathbb{R}^3$ is projected forward under the transition matrix $\mathbf{F} = \text{diag}(1, \rho, 1)$:

$$\hat{\mathbf{z}}_{t|t-1} = \mathbf{F}\hat{\mathbf{z}}_{t-1|t-1}, \qquad \mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^T + \mathbf{Q} \tag{21}$$

where $\mathbf{Q} = \text{diag}(10^{-12}, \sigma_M^2, \sigma_R^2)$. Because $q_\beta^2 \approx 0$, $\hat{\beta}_{t|t-1} \equiv \beta_0$.

**Step 3: Analytical Measurement Update.** Observing response price $X_{2,t}$ with exogenous driver $X_{1,t}$ via $\mathbf{H}_t = [X_{1,t}, 1, 1]$:

$$e_t = X_{2,t} - \mathbf{H}_t\hat{\mathbf{z}}_{t|t-1} = X_{2,t} - (\beta_0 X_{1,t} + \hat{M}_{t|t-1} + \hat{R}_{t|t-1}) \tag{22}$$

$$S_t = \mathbf{H}_t\mathbf{P}_{t|t-1}\mathbf{H}_t^T + R_{\text{obs}}, \qquad \mathbf{K}_t = \frac{\mathbf{P}_{t|t-1}\mathbf{H}_t^T}{S_t} \tag{23}$$

$$\hat{\mathbf{z}}_{t|t} = \hat{\mathbf{z}}_{t|t-1} + \mathbf{K}_t e_t, \qquad \mathbf{P}_{t|t} = (\mathbf{I}_3 - \mathbf{K}_t\mathbf{H}_t)\mathbf{P}_{t|t-1} \tag{24}$$

**Step 4: Signal Generation on Isolated $M_t$.** The trading signal is constructed strictly on the mean-reverting state estimate $\hat{M}_{t|t}$, normalized by its rolling or expanding standard deviation:

$$z_t = \frac{\hat{M}_{t|t} - \mu_M(W)}{\sigma_M(W)} \tag{25}$$

**Step 5: Dollar-Neutral Execution.** Positions are taken with static dollar-neutral weights:

$$w_2 = \frac{1}{1 + |\beta_0|}, \qquad w_1 = \frac{\beta_0}{1 + |\beta_0|} \tag{26}$$

entering when $|z_t| \geq \tau_{\text{open}}$ (production benchmark $\tau_{\text{open}} = 1.0\sigma$, with $1.5\sigma$ evaluated as a conservative alternative) and liquidating flat upon mean-crossing ($z_t = 0.0$). A strict 10 bps transaction friction is deducted on both legs upon every rebalance.

### 7.2 Method 2: Dynamic-PCI (Time-Varying Kalman Cointegrating Vector)

Dynamic-PCI resolves the vulnerability of static hedge ratios to multi-year corporate balance sheet drift by modeling $\beta_t$ as a stochastic random walk jointly with $M_t$ and $R_t$.

**Step 1: 5-Parameter Joint Maximum Likelihood Estimation.** On formation data, the parameter vector $\boldsymbol{\theta}_{\text{dyn}} = [\beta_0, \rho, \ln \sigma_M, \ln \sigma_R, \ln q_\beta]$ is optimized jointly via 5D prediction-error MLE, allowing $q_\beta > 0$ to be freely estimated from market returns.

**Step 2: Exogenous Driver Formulation.** The price $X_{1,t}$ enters the observation equation dynamically via $\mathbf{H}_t = [X_{1,t}, 1, 1]$. This formulation avoids inflating the state space to 4 latent dimensions, ensuring mathematical well-posedness and rapid convergence.

**Step 3: Joint 3-Way State Filtering.** The Kalman filter dynamically splits the prediction innovation $e_t = X_{2,t} - \hat{\beta}_{t|t-1} X_{1,t} - \hat{M}_{t|t-1} - \hat{R}_{t|t-1}$ across all three latent states through the analytical gain vector $\mathbf{K}_t = [K_{\beta,t}, K_{M,t}, K_{R,t}]^T$:

$$\hat{\beta}_{t|t} = \hat{\beta}_{t|t-1} + K_{\beta,t} e_t, \quad \hat{M}_{t|t} = \hat{M}_{t|t-1} + K_{M,t} e_t, \quad \hat{R}_{t|t} = \hat{R}_{t|t-1} + K_{R,t} e_t \tag{27}$$

**Step 4: Continuous Adaptive Hedge-Ratio Rebalancing.** Portfolio dollar-neutral weights dynamically adjust at every discrete bar using the posterior hedge ratio $\hat{\beta}_{t|t}$:

$$w_{2,t} = \frac{1}{1 + |\hat{\beta}_{t|t}|}, \qquad w_{1,t} = \frac{\hat{\beta}_{t|t}}{1 + |\hat{\beta}_{t|t}|} \tag{28}$$

**Step 5: Pure Mean-Reversion Execution.** Trading execution triggers exclusively from the standardized $M_t$ state, completely decoupling alpha generation from the structural drift $R_t$ and the evolving trend of $\beta_t$.

### 7.3 Method 3: Regime-Gated Dynamic-PCI (Dual Volatility & Half-Life Gate)

Regime-Gated Dynamic-PCI augments Method 2 with a real-time, online risk-control gate designed to halt trading during structural cointegration breakdowns or market volatility blowouts.

**Step 1: Causal Online State Estimation.** At bar $t$, posterior states $(\hat{\beta}_t, \hat{M}_t, \hat{R}_t)$ are estimated via Dynamic-PCI using only historical observations $\mathcal{F}_t$.

**Step 2: Causal Volatility Blowout Gate Evaluation.** The filter measures the ratio of short-term spread volatility (20-day rolling standard deviation) to long-term cumulative volatility:

$$gate_{\text{vol},t} = \mathbb{I}\left(\frac{\text{std}(\hat{M}_{t-20:t})}{\text{std}(\hat{M}_{1:t})} \leq 3.0\right) \tag{29}$$

If short-term spread volatility expands beyond $3\times$ baseline, $gate_{\text{vol},t} = 0$, indicating an anomalous market dislocation.

**Step 3: Rolling AR(1) Mean-Reversion Half-Life Gate.** On a rolling 60-day window of estimated states $\hat{M}_{t-60:t}$, an OLS autoregression is fit:

$$\Delta\hat{M}_s = \kappa(\mu - \hat{M}_{s-1}) + \epsilon_s, \quad s \in [t-60, t] \tag{30}$$

The empirical rolling half-life is computed as $HL_{\text{roll}} = \frac{\ln 2}{\kappa}$. If $\kappa \leq 0$ (explosive / unit-root path) or $HL_{\text{roll}} > 126$ trading days ($> 6$ months), mean-reversion has broken down:

$$gate_{\text{AR1},t} = \mathbb{I}\left(\kappa > 0 \quad \text{and} \quad HL_{\text{roll}} \leq 126.0\right) \tag{31}$$

**Step 4: Composite Binary Risk-Off Filter.** The master execution gate is defined as:

$$gate_t = gate_{\text{vol},t} \times gate_{\text{AR1},t} \in \{0, 1\} \tag{32}$$

**Step 5: Liquidation & Entry Prohibition Protocol.**

- When $gate_t = 1$: Trading rules execute normally according to $z(\hat{M}_t)$.
- When $gate_t = 0$: If an active position is currently held, it is **immediately closed to flat cash** ($pos_t = 0$), paying transaction costs to prevent holding an unhedged or diverging spread. If flat, all new entry orders are strictly suppressed.

## 7.4 Method 4: Neural KalmanBOT (KalmanNet Recurrent Architecture & Training)

Neural KalmanBOT replaces the closed-form analytical Kalman gain matrix $\mathbf{K}_t$ with a recurrent neural network (**KalmanNetGain**) trained end-to-end to maximize risk-adjusted strategy performance under market frictions.

### 7.4.1 Motivation: Overcoming Gaussian Assumptions

Analytical Kalman filtering assumes that measurement errors $v_t$ and process noises $\mathbf{w}_t$ are strictly Gaussian and that true covariances $(\mathbf{Q}, \mathbf{R}_{\text{obs}})$ are perfectly known. In real asset markets, asset returns exhibit heavy tails, volatility clustering, and conditional heteroskedasticity. Under these conditions, the analytical gain $\mathbf{K}_t$ overreacts to outlier price spikes or underreacts to regime shifts. KalmanNet learns an adaptive gain mapping directly from data.

### 7.4.2 Neural Network Architecture (Layer-by-Layer)

The architecture of `KalmanNetGain` in `neural_kalman_pci.py` is structured as follows:

1. **Input Feature Engineering ($\mathbf{u}_t \in \mathbb{R}^{m+n}$):** In the canonical KalmanNet framework (Revach et al., 2022), the network receives two scale-invariant feature vectors at each step $t$:

$$\tilde{\mathbf{e}}_t = \frac{\mathbf{e}_t}{\|\mathbf{e}_t\|_2 + \epsilon} \in \mathbb{R}^n, \qquad \Delta\tilde{\mathbf{z}}_{t-1} = \frac{\hat{\mathbf{z}}_{t-1|t-1} - \hat{\mathbf{z}}_{t-1|t-2}}{\|\hat{\mathbf{z}}_{t-1|t-1} - \hat{\mathbf{z}}_{t-1|t-2}\|_2 + \epsilon} \in \mathbb{R}^m \tag{33}$$

   where $m$ is the state dimension and $n$ is the observation dimension. The input feature vector is concatenated as $\mathbf{u}_t = [\tilde{\mathbf{e}}_t, \Delta\tilde{\mathbf{z}}_{t-1}]^T \in \mathbb{R}^{m+n}$.

   *Mathematical Correction on Prior State Propagation:* Notice that the candidate feature $\hat{\mathbf{z}}_{t|t-1} - \mathbf{F}\hat{\mathbf{z}}_{t-1|t-1}$ is **identically zero** ($\mathbf{0}$) by definition of the state transition prediction step ($\hat{\mathbf{z}}_{t|t-1} \equiv \mathbf{F}\hat{\mathbf{z}}_{t-1|t-1}$ in any linear time-invariant state-space model). Hence, KalmanNet's feature $F_4$ strictly utilizes the *previous step's update* (posterior minus prior $\hat{\mathbf{z}}_{t-1|t-1} - \hat{\mathbf{z}}_{t-1|t-2} = \mathbf{K}_{t-1}\mathbf{e}_{t-1}$) to track state innovation trajectory.

   *Addressing Scalar Observation Normalization:* When observation $y_t$ is a scalar ($n = 1$), naive L2 normalization $\frac{e_t}{|e_t|+\epsilon}$ reduces directly to $\text{sign}(e_t)$, discarding the vital magnitude of the price dislocation. In our production pipeline, pricing scale awareness is preserved by utilizing the coupled 2D observation vector $\mathbf{y}_t = [X_{2,t}, X_{1,t}]^T \in \mathbb{R}^2$ (preserving the angle and relative magnitude between follower spread error and driver price) and concatenating prior state estimates $\hat{\mathbf{z}}_{t|t-1}$.

2. **Layer 1: Input Feedforward Projection ($\mathbb{R}^{m+n} \to \mathbb{R}^{H_1}$):** A dense linear layer projects $\mathbf{u}_t$ to hidden representation $H_1 = \min(48, (m+n)\times 8)$ ($H_1 = 32$ for the reduced 3-state exogenous model, or $H_1 = 48$ for the 4-state joint system):

$$\mathbf{h}_t^{(1)} = \text{ReLU}\left(\mathbf{W}_1\mathbf{u}_t + \mathbf{b}_1\right) \in \mathbb{R}^{H_1} \tag{34}$$

3. **Layer 2: Recurrent Memory Core (Gated Recurrent Unit - GRU):** A single-layer GRU with input size $H_1$ and hidden state size $d_{\text{hidden}} = 32$—totaling exactly 6,336 trainable parameters for $H_1 = 32$ ($3 \times [32 \times 32 + 32 \times 32 + 2 \times 32]$) and 7,872 parameters for $H_1 = 48$ ($3 \times [48 \times 32 + 32 \times 32 + 2 \times 32]$)—tracks multi-period temporal dependencies across consecutive innovations:

$$\mathbf{r}_t = \sigma\left(\mathbf{W}_{ir}\mathbf{h}_t^{(1)} + \mathbf{W}_{hr}\mathbf{h}_{t-1} + \mathbf{b}_r\right) \tag{35}$$

$$\mathbf{z}_t^{\text{gru}} = \sigma\left(\mathbf{W}_{iz}\mathbf{h}_t^{(1)} + \mathbf{W}_{hz}\mathbf{h}_{t-1} + \mathbf{b}_z\right) \tag{36}$$

$$\mathbf{n}_t = \tanh\left(\mathbf{W}_{in}\mathbf{h}_t^{(1)} + \mathbf{b}_{in} + \mathbf{r}_t \odot (\mathbf{W}_{hn}\mathbf{h}_{t-1} + \mathbf{b}_{hn})\right) \tag{37}$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t^{\text{gru}}) \odot \mathbf{n}_t + \mathbf{z}_t^{\text{gru}} \odot \mathbf{h}_{t-1} \in \mathbb{R}^{32} \tag{38}$$

   The recurrent hidden state $\mathbf{h}_t$ serves as a memory buffer capturing whether the spread is experiencing transient noise or entering a prolonged structural trend.

4. **Layer 3: Hidden Projection Layer ($\mathbb{R}^{32} \to \mathbb{R}^{H_2}$):** A second dense layer compresses the recurrent state down to $H_2 = \min(24, \max(8, 2mn))$:

$$\mathbf{h}_t^{(2)} = \text{ReLU}\left(\mathbf{W}_2\mathbf{h}_t + \mathbf{b}_2\right) \in \mathbb{R}^{H_2} \tag{39}$$

5. **Layer 4: Output Gain Head ($\mathbb{R}^{H_2} \to \mathbb{R}^{m\times n}$):** The final linear head maps $H_2$ to the flattened Kalman gain matrix:

$$\mathbf{k}_t = \mathbf{W}_{\text{out}}\mathbf{h}_t^{(2)} + \mathbf{b}_{\text{out}} \implies \mathbf{K}_t^{\text{NN}} \in \mathbb{R}^{m\times n} \tag{40}$$

**Conservative Zero-Initialization:** Crucially, weights and biases of Layer 4 are strictly initialized to zero ($\mathbf{W}_{\text{out}} = \mathbf{0}, \mathbf{b}_{\text{out}} = \mathbf{0}$). Consequently, prior to training, $\mathbf{K}_t^{\text{NN}} \equiv \mathbf{0}$. The model starts by conservatively trusting prior state predictions $\hat{\mathbf{z}}_{t|t-1}$ without erratic initialization jumps, learning gain corrections incrementally.

#### 7.4.3 Posterior State Update & Clamping

Given the learned gain $\mathbf{K}_t^{\text{NN}}$, the posterior state vector is updated and numerically clamped:

$$\hat{\mathbf{z}}_{t|t} = \text{clip}\left(\hat{\mathbf{z}}_{t|t-1} + \mathbf{K}_t^{\text{NN}} e_t, -50.0, 50.0\right) \tag{41}$$

Clamping prevents feedback explosion during recurrent backpropagation.

#### 7.4.4 Two-Stage Training Curriculum

KalmanNet is trained under a two-stage curriculum:

**Stage 1: Observation MSE Pre-Training (15 Epochs).** Network parameters are pre-trained to minimize one-step prediction error on the formation response series $X_{2,t}$:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{T}\sum_{t=1}^{T}\left(X_{2,t} - \mathbf{H}_t\hat{\mathbf{z}}_{t|t-1}\right)^2 \tag{42}$$

This anchors the network in physical tracking before strategy optimization.

**Stage 2: Differentiable Sharpe Ratio Fine-Tuning via BPTT (15 Epochs).** Stage 2 directly aligns the Kalman gain with trading profitability under transaction frictions:

1. Continuous position proxy: $\widetilde{\text{pos}}_t = -\tanh\left(\frac{z(\hat{M}_t)}{\tau_{\text{soft}}}\right)$, where $\tau_{\text{soft}} = 1.0$.
2. Strategy net daily returns accounting for 10 bps transaction costs:

$$R_t^{\text{strat}} = \widetilde{\text{pos}}_{t-1}\left(w_{2,t-1}\frac{\Delta X_{2,t}}{X_{2,t-1}} - w_{1,t-1}\frac{\Delta X_{1,t}}{X_{1,t-1}}\right) - c\cdot|\widetilde{\text{pos}}_t - \widetilde{\text{pos}}_{t-1}| \tag{43}$$

3. Objective function minimized via Backpropagation Through Time (BPTT):

$$\mathcal{L}_{\text{Sharpe}} = -\frac{\mathbb{E}[R_t^{\text{strat}}]}{\sqrt{\text{Var}(R_t^{\text{strat}}) + 10^{-8}}} \times \sqrt{252} \tag{44}$$

Gradient norms are clipped at $\|\mathbf{g}\|_2 \leq 1.0$, and the recurrent hidden state $\mathbf{h}_{t-1}$ is detached across sequence batch boundaries to guarantee numerical stability.

### 7.5 Comprehensive Architectural Comparison Matrix

Table 2 summarizes the structural differences across all four evaluated frameworks.

Table 2: Architectural and Mathematical Comparison of Evaluated Pairs Trading Frameworks

| Feature | Static-PCI | Dynamic-PCI | Regime-Gated | Neural KalmanBOT |
|---|---|---|---|---|
| **Hedge Ratio ($\beta_t$)** | Static $\beta_0$ (fixed at MLE). | Random walk $\beta_t = \beta_{t-1} + w_\beta$. | Random walk $\beta_t = \beta_{t-1} + w_\beta$. | Random walk $\beta_t = \beta_{t-1} + w_\beta$. |
| **Filter Gain ($\mathbf{K}_t$)** | Closed-form analytical Kalman gain. | Closed-form analytical Kalman gain. | Closed-form analytical Kalman gain. | Learned GRU network (`KalmanNetGain`). |
| **Spread Model** | $M_t \sim \text{AR}(1), R_t \sim I(1)$. | $M_t \sim \text{AR}(1), R_t \sim I(1)$. | $M_t \sim \text{AR}(1), R_t \sim I(1)$. | $M_t \sim \text{AR}(1), R_t \sim I(1)$. |
| **Risk Control** | Pre-trade screening only. | Pre-trade screening only. | **Online Dual Gate** (Vol ratio + rolling $HL$). | **Learned gain attenuation** via Sharpe loss. |
| **Cost Awareness** | Execution accounting (10 bps). | Execution accounting (10 bps). | Execution accounting (10 bps). | **Embedded in loss function** ($c = 10$ bps). |
| **Primary Strength** | Low turnover on true stationary spreads. | Tracks evolving corporate capital structures. | Caps catastrophic drawdowns during breaks. | Reverses severe structural losses into profits. |

# 8 Data Universe & Empirical Benchmark Results

The benchmark is evaluated across an expanded 14-pair cross-asset universe spanning 12 years (2009–2021), incorporating canonical equities, commodities/energy, and the newly validated passing pairs (`MS--GS`, `V--MA`, `HD--LOW`):

- **Formation Period (In-Sample):** Exactly 2,014 trading days (2009-09-01 to 2017-08-30).
- **Trading Period (Out-of-Sample):** Exactly 1,007 trading days (2017-08-31 to 2021-08-31).

Table 3: Production Out-of-Sample Performance Summary Across 14 Pairs (1,007 Trading Days, Cost = 10 bps per half-turn)

| Pair | Model | TotRet | AnnRet | Sharpe | MaxDD | Key Empirical Impact |
|---|---|---|---|---|---|---|
| **AAP–IBM** | Static-PCI | −7.27% | −1.87% | 0.01 | −44.52% | Non-stationary downward drift |
| *[REJ]* | Dynamic-PCI | −24.59% | −6.82% | −0.27 | −43.55% | Over-adjusts to persistent drift |
| | Regime-Gated | −28.72% | −8.12% | −0.38 | −45.44% | Whip-saw stops |
| | Neural KalmanBOT | −30.73% | −8.78% | −0.41 | −47.23% | Structural breakdown trend |
| **EWA–EWC** | **Static-PCI** | **+1.89%** | **+0.47%** | **0.11** | **−9.30%** | **Cointegrated resource pair; lowest MaxDD** |
| *[PASS]* | Dynamic-PCI | −5.16% | −1.32% | −0.18 | −13.21% | Slight negative drift |
| | Regime-Gated | −3.69% | −0.94% | −0.16 | −14.31% | Gated out during small dislocations |
| | Neural KalmanBOT | −2.90% | −0.73% | −0.08 | −14.45% | Comparable to dynamic baseline |
| **NKE–AES** | Static-PCI | −64.40% | −22.78% | −0.69 | −69.47% | Severe structural breakdown loss |
| *[REJ]* | Dynamic-PCI | −15.97% | −4.26% | −0.06 | −52.64% | Better tracking than static |
| | Regime-Gated | −10.34% | −2.69% | −0.01 | −46.71% | Partial loss reduction |

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**Table 3 continued from previous page**

| Pair | Model | TotRet | AnnRet | Sharpe | MaxDD | Key Empirical Impact |
|---|---|---|---|---|---|---|
| | **Neural KalmanBOT** | **+45.70%** | **+9.88%** | **0.55** | **−23.04%** | **Major turnaround:** +45.7% **return, MaxDD** −23.0% |
| **MRK–VZ** | **Static-PCI** | **+37.72%** | **+8.34%** | **0.82** | **−7.00%** | **Strongest stationary alpha across benchmark** |
| *[REJ]* | Dynamic-PCI | −0.25% | −0.06% | 0.05 | −15.13% | Near break-even |
| | Regime-Gated | −5.74% | −1.47% | −0.18 | −13.19% | Filter lag reduces return |
| | **Neural KalmanBOT** | **+40.23%** | **+8.83%** | **0.75** | **−10.38%** | **Captures substantial stationary alpha** (+40.2%) |
| **GS–CVX** | Static-PCI | +19.56% | +4.57% | 0.40 | −24.30% | Profitable cyclical dislocation |
| *[REJ]* | Dynamic-PCI | −1.80% | −0.45% | 0.03 | −26.48% | Modest loss |
| | Regime-Gated | −0.13% | −0.03% | 0.04 | −16.50% | Flat, reduced MaxDD |
| | **Neural KalmanBOT** | **+32.39%** | **+7.27%** | **0.41** | **−43.86%** | **Highest cumulative gain** (+32.4%) |
| **KO–PEP** | Static-PCI | −9.11% | −2.36% | −0.27 | −23.18% | Non-stationary spread drift |
| *[REJ]* | Dynamic-PCI | −2.83% | −0.72% | −0.04 | −14.33% | Improved over static |
| | Regime-Gated | −6.77% | −1.74% | −0.36 | −11.51% | Lower drawdown |
| | **Neural KalmanBOT** | **+3.92%** | **+0.97%** | **0.15** | **−10.18%** | **Best performing model, positive Sharpe** (0.15) |
| **XOM–CVX** | Static-PCI | +20.54% | +4.79% | 0.59 | −9.83% | Good stationary energy alpha |
| *[REJ]* | Dynamic-PCI | −11.99% | −3.15% | −0.23 | −32.80% | Excess beta sensitivity |
| | Regime-Gated | −4.07% | −1.03% | −0.16 | −15.15% | Mitigates dynamic loss |
| | **Neural KalmanBOT** | **+40.11%** | **+8.81%** | **0.98** | **−11.12%** | **Highest Sharpe** (0.98) **and return** (+40.1%) |
| **GLD–SLV** | Static-PCI | −25.28% | −7.03% | −1.11 | −25.51% | Heavy metals dislocation loss |
| *[REJ]* | **Dynamic-PCI** | **+13.24%** | **+3.16%** | **0.46** | **−16.75%** | **Beta tracking adapts to macro drift** |
| | **Regime-Gated** | **−2.73%** | **−0.69%** | **−0.25** | **−6.48%** | **Shields capital; MaxDD cut from** −25.5% **to** −6.5% |
| | Neural KalmanBOT | −27.54% | −7.75% | −1.26 | −27.71% | Unmodeled structural break |
| **XLE–XOP** | Static-PCI | −14.52% | −3.85% | −0.43 | −20.35% | E&P divergence |
| *[REJ]* | Dynamic-PCI | −10.56% | −2.75% | −0.07 | −40.64% | High volatility |
| | **Regime-Gated** | **−0.58%** | **−0.15%** | **−0.68** | **−0.58%** | **97% drawdown reduction (MaxDD** −0.58%) |
| | Neural KalmanBOT | −6.12% | −1.57% | −0.13 | −16.91% | Reduced drawdown vs dynamic |
| **XLE–OIH** | Static-PCI | +15.57% | +3.69% | 0.35 | −19.73% | Profitable recovery |
| *[PASS-PCI]* | **Dynamic-PCI** | **+16.47%** | **+3.89%** | **0.28** | **−40.06%** | **Highest return, but** −40.1% **MaxDD** |
| | Regime-Gated | −31.92% | −9.17% | −0.75 | −39.27% | Adverse gate exits |
| | Neural KalmanBOT | −5.76% | −1.47% | −0.12 | **−13.57%** | **Lowest MaxDD** (−13.6% **vs** −40.1%) |
| **USO–XLE** | Static-PCI | −23.31% | −6.43% | −0.27 | −44.09% | Contango/roll-yield breakdown |
| *[REJ]* | Dynamic-PCI | −34.82% | −10.16% | −0.36 | −56.53% | Deep commodity trend drawdown |
| | Regime-Gated | −37.56% | −11.12% | −0.42 | −56.53% | Persistent non-stationarity |
| | **Neural KalmanBOT** | **−15.46%** | **−4.12%** | **−0.10** | **−29.86%** | **Drawdown cut by nearly half** (−29.9%) |
| **MS–GS** | **Static-PCI** | **+16.37%** | **+3.87%** | **0.50** | **−9.37%** | **Premier banking pair, passes strict screen** |
| *[PASS]* | Dynamic-PCI | −16.77% | −4.49% | −0.42 | −22.93% | Fast beta drift whipsaw |
| | Regime-Gated | −17.55% | −4.71% | −0.66 | −23.65% | Conservative gate timing |
| | Neural KalmanBOT | −11.65% | −3.05% | −0.22 | −26.21% | Controlled downside |

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**Table 3 continued from previous page**

| Pair | Model | TotRet | AnnRet | Sharpe | MaxDD | Key Empirical Impact |
|---|---|---|---|---|---|---|
| **V–MA** | Static-PCI | −8.93% | −2.31% | −0.46 | −12.33% | Payment network duopoly |
| *[PASS-PCI]* | Dynamic-PCI | −8.84% | −2.29% | −0.54 | −14.30% | Fast mean reversion ($HL$ = 1.0d) |
| | Regime-Gated | −8.84% | −2.29% | −0.54 | −14.30% | Identical to dynamic |
| | Neural KalmanBOT | −14.04% | −3.72% | −0.69 | −15.27% | Conservative sizing |
| **HD–LOW** | Static-PCI | −21.42% | −5.85% | −0.52 | −26.74% | Home improvement retail duopoly |
| *[PASS-PCI]* | Dynamic-PCI | −11.90% | −3.12% | −0.33 | −20.05% | Improved over static |
| | Regime-Gated | −11.67% | −3.06% | −0.32 | −19.84% | Dual gate active |
| | **Neural KalmanBOT** | **+6.83%** | **+1.67%** | **0.23** | **−18.56%** | **Reverses** −21.4% **loss into** +6.8% **gain!** |

# 9 Screening Review, Formation Window Dynamics, and Execution Rule Optimization

## 9.1 Revisiting Formation Screening: Theory, Diagnosis, and Criteria Relaxation

### 9.1.1 Theoretical Rationale of the Screening Engine

The primary objective of the formation screening stage is **capital preservation via structural validation**. In statistical arbitrage, pair trading operates on the premise that the spread between two economically linked assets exhibits stationary, mean-reverting behavior. Under the Partial Cointegration (Clegg & Krauss, 2018) state-space decomposition:

$$X_{2,t} - \beta_t X_{1,t} = M_t + R_t \tag{45}$$

the observed residual is partitioned into a mean-reverting component $M_t = \rho M_{t-1} + \varepsilon_{M,t}$ ($|\rho| < 1$) and a permanent random walk drift $R_t = R_{t-1} + \varepsilon_{R,t}$. The screening engine enforces four fundamental quantitative objectives:

1. **Preventing Divergent Capital Loss:** If $\sigma_R^2 \gg \sigma_M^2$ or $\rho \geq 1.0$, spread deviations do not mean-revert. Trading such pairs results in open-ended drawdown as positions accumulate catastrophic losses on non-stationary trends.
2. **Ensuring Signal Quality ($R^2_{\mathbf{MR}} \geq 15\%$):** The mean-reverting variance proportion $R^2_{\text{MR}} = \frac{\text{Var}(\Delta M_t)}{\text{Var}(\Delta M_t)+\text{Var}(\Delta R_t)} = \frac{2\sigma_M^2}{2\sigma_M^2+(1+\rho)\sigma_R^2}$ (Clegg & Krauss, 2018) guarantees that observed dislocations represent exploitable economic inefficiencies rather than pure random-walk noise.
3. **Guaranteeing Practical Holding Horizons ($HL \leq 252$ days):** The half-life of mean-reversion $HL = -\frac{\ln 2}{\ln|\rho|}$ must be finite to ensure capital turns over within an operational fiscal horizon, rejecting multi-year capital lockup.
4. **Parameter Stability ($\rho < 0.999$):** Rejecting boundary solutions where the transition matrix approaches a pure unit root.

### 9.1.2 Diagnosis of the Legacy Screening Failure Mode

In the baseline pipeline, 10 out of 11 canonical pairs failed the strict diagnostic screen (only `EWA--EWC` passed). A forensic re-examination revealed a major theoretical contradiction in the legacy testing methodology:

- **Static Spread vs. Dynamic Spread:** The legacy Augmented Dickey–Fuller (ADF) test was computed on the raw static spread $s_t^{\text{static}} = X_{2,t} - \beta_0 X_{1,t}$ using a single fixed hedge ratio $\beta_0$ over an 8-year formation period (2,014 trading days). Over an 8-year span, the true economic hedge ratio $\beta_t$ drifts due to evolving leverage, dividend policies, and operational changes. Any cumulative drift in $\beta_t$ imparts an apparent non-stationary trend to $s_t^{\text{static}}$, causing the static ADF test to fail catastrophically ($p = 0.9641$ for `KO--PEP`, $p = 0.9874$ for `HD--LOW`).
- **Theoretical Contradiction with Partial Cointegration:** Partial cointegration explicitly posits that the spread contains a permanent drift component $R_t$ ($\sigma_R^2 > 0$). By definition, the sum $M_t + R_t$ is **asymptotically non-stationary**. Demanding that the raw spread pass a strict ADF unit-root test ($p \leq 0.10$) tests for *full static cointegration*, contradicting the very purpose of deploying a partial cointegration filter.

### 9.1.3 Relaxed & PCI-Aware Screening Framework

To align empirical screening with state-space theory, we formulated two relaxed screening tiers:

1. **Relaxed Tier 1:** $R^2_{\text{MR}} \geq 15\%$, $HL \leq 365$ days, Static $p_{\text{ADF}} \leq 0.20$, and $\rho < 0.999$.
2. **PCI-Aware Tier:** $R^2_{\text{MR}} \geq 10\%$, $HL \leq 365$ days, $\rho < 0.999$, and **ADF Stationarity verified directly on the filtered mean-reverting state** $p_{\text{ADF}}(M_t) \leq 0.05$ (or Static $p_{\text{ADF}} \leq 0.25$).

### 9.1.4 Empirical Screening Results Across an Expanded 21-Pair Universe

We evaluated the screening framework across an expanded universe of 21 cross-asset pairs spanning equity duopolies, mega-cap tech, money-center banks, industrial logistics, and ETFs over the 2009–2017 formation window (Table 4).

Table 4: Screening Diagnostic Statistics Across Expanded 21-Pair Universe (Formation: 2009–2017, 2,014 Days)

| Pair | $R^2_{\text{MR}}$ | Half-Life | $\rho$ | $p^{\text{static}}_{\text{ADF}}$ | $p^{M_t}_{\text{ADF}}$ | Strict | Relaxed | PCI-Aware |
|---|---|---|---|---|---|---|---|---|
| `KO--PEP` | 36.6% | 692.8d | 0.999 | 0.9641 | 0.4317 | Fail | Fail | Fail |
| `EWA--EWC` | 78.6% | 136.4d | 0.995 | 0.0439 | 0.1430 | **Pass** | **Pass** | **Pass** |
| `XOM--CVX` | 80.4% | 692.8d | 0.999 | 0.2432 | 0.7182 | Fail | Fail | Fail |
| `MRK--VZ` | 51.7% | 692.8d | 0.999 | 0.4882 | 0.6429 | Fail | Fail | Fail |
| `GS--CVX` | 10.8% | 498.5d | 0.999 | 0.1736 | 0.3103 | Fail | Fail | Fail |
| `GLD--SLV` | 41.0% | 692.8d | 0.999 | 0.3142 | 0.8577 | Fail | Fail | Fail |
| `XLE--XOP` | 13.4% | 692.8d | 0.999 | 0.9383 | 0.6295 | Fail | Fail | Fail |
| `XLE--OIH` | 62.5% | 81.5d | 0.992 | 0.6900 | 0.0027 | Fail | Fail | **Pass** |
| `USO--XLE` | 14.2% | 4.2d | 0.848 | 0.3329 | $< 10^{-4}$ | Fail | Fail | **Pass** |
| `V--MA` | 33.9% | 1.0d | 0.488 | 0.7114 | $< 10^{-4}$ | Fail | Fail | **Pass** |
| `HD--LOW` | 50.9% | 2.0d | 0.707 | 0.9874 | $< 10^{-4}$ | Fail | Fail | **Pass** |
| `MSFT--AAPL` | 46.7% | 692.8d | 0.999 | 0.7779 | 0.2585 | Fail | Fail | Fail |
| `GOOGL--META` | 23.3% | 692.8d | 0.999 | 0.9989 | 0.7248 | Fail | Fail | Fail |
| `JPM--BAC` | 30.9% | 425.5d | 0.998 | 0.6755 | 0.3322 | Fail | Fail | Fail |
| `MS--GS` | **90.8%** | **68.5d** | **0.990** | **0.0490** | **0.0005** | **Pass** | **Pass** | **Pass** |
| `CAT--DE` | 60.9% | 692.8d | 0.999 | 0.3279 | 0.7344 | Fail | Fail | Fail |
| `UPS--FDX` | 63.0% | 692.8d | 0.999 | 0.5019 | 0.3983 | Fail | Fail | Fail |
| `WMT--TGT` | 100.0% | 511.5d | 0.999 | 0.3891 | 0.4767 | Fail | Fail | Fail |
| `JNJ--PG` | 53.1% | 692.8d | 0.999 | 0.0003 | 0.7007 | Fail | Fail | Fail |
| `SPY--IVV` | 37.2% | 2.8d | 0.782 | 0.6055 | $< 10^{-4}$ | Fail | Fail | **Pass** |
| `XLK--QQQ` | 97.7% | $\infty$ | $-0.879$ | 0.7929 | $< 10^{-4}$ | Fail | Fail | Fail |

**Key Screening Discoveries:**

1. **Discovery of `MS--GS` (Morgan Stanley vs. Goldman Sachs):** `MS--GS` passes **ALL 4 STRICT CRITERIA** ($R^2_{\text{MR}} = 90.8\%$, Half-Life = 68.5 days, $\rho = 0.9899$, and Static $p_{\text{ADF}} = 0.0490 < 0.05$). The filtered component $M_t$ is stationary at $p = 0.0005$. This confirms `MS--GS` as a new, production-grade candidate pair with massive transient variance and fast mean reversion.

2. **PCI-Aware Passes (`V--MA`, `HD--LOW`, `SPY--IVV`, `XLE--OIH`):** When evaluated under the PCI-Aware test, these pairs reveal rapid mean reversion on the filtered state ($HL = 1.0$d for `V--MA`, 2.0d for `HD--LOW`, 2.8d for `SPY--IVV`, and 81.5d for `XLE--OIH`), confirming that their apparent non-stationarity under classical ADF was caused entirely by slow drift in hedge ratio $\beta_t$ rather than loss of economic linkage.

### 9.2 Formation Window Duration Dynamics (In-Sample vs. Out-of-Sample)

To understand how the choice of formation length affects model estimation and out-of-sample trading performance, we systematically varied the in-sample formation window across 1 Year (252d), 2 Years (504d), 3 Years (756d), 5 Years (1,260d), and 8 Years (2,014d), evaluating the **Dynamic-PCI** strategy over a fixed out-of-sample trading period of 1,007 trading days (2017-08-31 to 2021-08-31).

Table 5: Impact of Formation Window Length on In-Sample Parameters and Out-of-Sample Performance (Dynamic-PCI Model, Cost = 10 bps)

| Pair | Formation Window | Days | $R^2_{\text{MR}}$ | Half-Life | $p_{\text{ADF}}$ | OOS Return | Sharpe | MaxDD |
|---|---|---|---|---|---|---|---|---|
| `KO--PEP` | 1 Year (252d) | 253 | 60.3% | 20.6d | 0.4353 | +0.92% | 0.07 | −13.09% |
| | **2 Years (504d)** | **505** | **41.2%** | **692.8**d | **0.7822** | **+18.34%** | **0.60** | **−5.78%** |
| | 3 Years (756d) | 756 | 70.6% | 10.5d | 0.7394 | −10.11% | −0.33 | −16.40% |
| | 5 Years (1260d) | 1258 | 51.5% | 4.2d | 0.8651 | −13.00% | −0.50 | −17.69% |
| | 8 Years (2014d) | 2014 | 36.6% | 692.8d | 0.9641 | −2.83% | −0.04 | −14.33% |
| `EWA--EWC` | 1 Year (252d) | 253 | 56.5% | 3.0d | 0.0918 | −24.35% | −1.13 | −26.10% |
| | 2 Years (504d) | 505 | 100.0% | 52.5d | 0.2794 | +0.21% | 0.04 | −9.63% |
| | 3 Years (756d) | 756 | 54.6% | 18.6d | 0.0651 | −7.81% | −0.29 | −17.70% |
| | 5 Years (1260d) | 1258 | 100.0% | 74.8d | 0.1065 | −4.93% | −0.15 | −14.34% |
| | 8 Years (2014d) | 2014 | 78.6% | 136.4d | 0.0439 | −5.16% | −0.18 | −13.21% |
| `V--MA` | 1 Year (252d) | 253 | 32.5% | 692.8d | 0.2316 | −1.73% | −0.05 | −8.10% |
| | 2 Years (504d) | 505 | 94.4% | 120.0d | 0.7599 | −6.73% | −0.28 | −8.77% |
| | 3 Years (756d) | 756 | 100.0% | 80.7d | 0.4485 | −10.11% | −0.38 | −10.11% |
| | 5 Years (1260d) | 1258 | 48.1% | 4.5d | 0.4897 | −7.67% | −0.38 | −10.39% |
| | 8 Years (2014d) | 2014 | 33.9% | 1.0d | 0.7114 | −8.84% | −0.54 | −14.30% |
| `HD--LOW` | 1 Year (252d) | 253 | 100.0% | 0.9d | 0.7007 | −19.30% | −0.75 | −20.00% |
| | 2 Years (504d) | 505 | 100.0% | $\infty$ | 0.8162 | −31.66% | −1.82 | −33.92% |
| | 3 Years (756d) | 756 | 100.0% | $\infty$ | 0.7692 | −37.28% | −1.85 | −39.65% |
| | 5 Years (1260d) | 1258 | 100.0% | 3.1d | 0.1792 | −22.63% | −0.67 | −27.00% |
| | 8 Years (2014d) | 2014 | 50.9% | 2.0d | 0.9874 | −11.90% | −0.33 | −20.05% |

**Quantitative Findings on Formation Window Length:**

1. **The 2-Year Empirical "Sweet Spot":** For `KO--PEP`, contracting the formation window from 8 years to 2 years (504 trading days) transformed an out-of-sample loss of −2.83% into a robust **+18.34**% gain with a Sharpe of **0.60** and a modest −5.78% maximum drawdown. A 2-year window provides sufficient data points (∼500) for L-BFGS-B MLE convergence while reflecting the *contemporary* economic relationship of the pair.

2. **Estimation Instability at 1 Year (252d):** A 1-year window is too brief for stable covariance estimation. In `EWA--EWC`, 1-year parameter estimation produced a catastrophic out-of-sample loss of −24.35% (Sharpe −1.13), driven by overfitting to localized short-term volatility shocks.

3. **Structural Aggregation Bias at 5–8 Years:** Over 8 years, parameter estimates suffer from structural aggregation over multiple distinct macroeconomic regimes (post-2008 recovery, zero-rate regimes, and energy cycles). The MLE optimizer forces $\rho$ toward the 0.999 boundary, inflating estimated half-lives to the upper cap (692.8d) and producing stale priors that degrade trading execution.

## 9.3 Comparative Analysis of Trade Execution Rules: Clegg & Krauss (2018) vs. Rolling Formulation

### 9.3.1 Specification of Execution Rules

In Section 4.2 of their seminal paper, Clegg & Krauss (2018) specify trade entry and exit based on the standardized spread dislocation. We compared four distinct execution formulations:

1. **Standard Production Rule (Rolling 80-Day Window, $\tau_{op} = 1.0$):** $z_t = \frac{M_t - \mu_{M,80}}{\sigma_{M,80}}$, entering at $|z_t| \geq 1.0$ and exiting at zero crossing ($z_t = 0$). This corresponds exactly to the Dynamic-PCI production benchmark in Table 3.

2. **Conservative Threshold Rule (Rolling 80-Day Window, $\tau_{op} = 1.5$):** Higher-conviction entry at $|z_t| \geq 1.5$ and exiting at equilibrium crossing ($z_t = 0$).

3. **Clegg & Krauss Expanding Window:** Standardizing $M_t$ using all available historical data up to time $t$, entering at $|z_t| \geq 1.0$ and exiting at equilibrium crossing ($z_t = 0$).

4. **Clegg & Krauss Half-Exit Rule:** Entering at $|z_t| \geq 1.0$ (or 1.5) and taking profit early when the dislocation reverts halfway to zero ($|z_t| \leq 0.5$).

5. **Theoretical Unconditional Z-Score:** Normalizing $M_t$ by the stationary standard deviation of the fitted AR(1) state:
$$z_t^{\text{uncond}} = \frac{M_t}{\sigma_M / \sqrt{1 - \rho^2}} \tag{46}$$
entering at $|z_t^{\text{uncond}}| \geq 1.0$ and closing at zero crossing.

**Trade Rule Comparison Verdict: Which Rule is Better?**

1. **Clegg & Krauss Expanding Z Decisively Beats the Rolling 80-Day Window:** On every tested pair, the C&K expanding window produced superior risk-adjusted returns (`MRK--VZ`: +16.44%, Sharpe 0.38 vs. +10.50%, Sharpe 0.29; `EWA--EWC`: +9.49%, Sharpe 0.41 vs. +1.70%, Sharpe 0.10).

2. **Why Rolling Windows Fail ("Dislocation Lag" Trap):** In a rolling 80-day window, when an asset experiences a sustained spread deviation lasting 40–60 days, the moving mean $\mu_{M,80}$ shifts toward the dislocated price level and the moving standard deviation inflates. Consequently, the calculated rolling $z_t$ drops back to zero *before the spread has actually reverted to fundamental equilibrium*, triggering premature exits at a loss. In contrast, the C&K expanding window anchors the long-term equilibrium, maintaining conviction until true economic reversion occurs.

3. **Theoretical Unconditional Z Delivers Exceptional Selectivity:** Theoretical Unconditional Z produced the highest Sharpe ratio across the entire study (**0.71** on `EWA--EWC` with only $-$**2.87**% MaxDD, and **0.35** on `KO--PEP` with +9.51% return). By normalizing $M_t$ by the exact MLE asymptotic state variance $\sigma_M / \sqrt{1 - \rho^2}$, it eliminates all rolling window distortions, executing only genuine statistical anomalies.

Table 6: Empirical Performance Comparison of Trade Execution Rules Across 4 Canonical Pairs (2017–2021 OOS)

| Pair | Execution Rule | Total Ret | Ann. Ret | Sharpe | MaxDD | Trades |
|---|---|---|---|---|---|---|
| KO--PEP | Standard ($\tau_{op} = 1.0$) [Table 3 benchmark] | −2.83% | −0.72% | −0.04 | −14.33% | 22 |
| | Conservative Threshold ($\tau_{op} = 1.5$) | −4.49% | −1.14% | −0.10 | −12.12% | 20 |
| | **C&K Expanding Z ($\tau_{op} = 1.0$)** | **+0.06%** | **+0.02%** | **0.04** | **−10.29%** | **15** |
| | C&K Half-Exit (Roll 80d, $\tau_{op} = 1.0, \tau_{cl} = 0.5$) | −1.50% | −0.38% | −0.00 | −14.34% | 21 |
| | C&K Half-Exit (Roll 80d, $\tau_{op} = 1.5, \tau_{cl} = 0.5$) | −3.11% | −0.79% | −0.06 | −12.13% | 19 |
| | **Theoretical Uncond-Z ($\tau_{op} = 1.0$)** | **+9.51%** | **+2.30%** | **0.35** | **−9.47%** | **4** |
| MRK--VZ | Standard ($\tau_{op} = 1.0$) [Table 3 benchmark] | −0.25% | −0.06% | 0.05 | −15.13% | 23 |
| | Conservative Threshold ($\tau_{op} = 1.5$) | +10.50% | +2.53% | 0.29 | −10.55% | 23 |
| | **C&K Expanding Z ($\tau_{op} = 1.0$)** | **+16.44%** | **+3.88%** | **0.38** | **−10.50%** | **14** |
| | C&K Half-Exit (Roll 80d, $\tau_{op} = 1.0, \tau_{cl} = 0.5$) | −7.06% | −1.82% | −0.10 | −20.47% | 19 |
| | C&K Half-Exit (Roll 80d, $\tau_{op} = 1.5, \tau_{cl} = 0.5$) | +2.04% | +0.51% | 0.10 | −11.66% | 19 |
| | Theoretical Uncond-Z ($\tau_{op} = 1.0$) | +6.30% | +1.54% | 0.29 | −6.69% | 2 |
| XOM--CVX | Standard ($\tau_{op} = 1.0$) [Table 3 benchmark] | −11.99% | −3.15% | −0.23 | −32.80% | 41 |
| | Conservative Threshold ($\tau_{op} = 1.5$) | −9.35% | −2.43% | −0.18 | −30.10% | 35 |
| | C&K Expanding Z ($\tau_{op} = 1.0$) | −5.14% | −1.31% | −0.05 | −30.41% | 17 |
| | **C&K Half-Exit (Roll 80d, $\tau_{op} = 1.5, \tau_{cl} = 0.5$)** | **+1.31%** | **+0.33%** | **0.08** | **−27.39%** | **29** |
| | Theoretical Uncond-Z ($\tau_{op} = 1.0$) | −3.12% | −0.79% | −0.01 | −30.41% | 1 |
| EWA--EWC | Standard ($\tau_{op} = 1.0$) [Table 3 benchmark] | −5.16% | −1.32% | −0.18 | −13.21% | 41 |
| | Conservative Threshold ($\tau_{op} = 1.5$) | +1.70% | +0.42% | 0.10 | −8.84% | 30 |
| | **C&K Expanding Z ($\tau_{op} = 1.0$)** | **+9.49%** | **+2.29%** | **0.41** | **−6.64%** | **27** |
| | C&K Half-Exit (Roll 80d, $\tau_{op} = 1.0, \tau_{cl} = 0.5$) | −5.86% | −1.50% | −0.19 | −12.96% | 35 |
| | **Theoretical Uncond-Z ($\tau_{op} = 1.0$)** | **+11.44%** | **+2.75%** | **0.71** | **−2.87%** | **5** |

4. **Role of the Half-Exit Rule on Trending Pairs:** On diverging spreads (XOM--CVX), the Half-Exit rule turned a −9.35% loss into a **+1.31%** profit. Exiting halfway limits exposure to the permanent random-walk drift $R_t$. However, on stationary pairs (MRK--VZ, EWA--EWC), exiting early cuts winners prematurely, forfeiting substantial reversion profit.

# 10 Failure Modes & Root-Cause Analyses

1. **Decoupled (Cascaded) Kalman Filtering:** Estimating $\beta_t$ in isolation assumes white-noise residuals, violating the fundamental partial cointegration structure where residuals contain autocorrelated $M_t$ and random-walk $R_t$. Furthermore, passing point estimates $\hat{\beta}_t$ into a second filter discards posterior covariance $P_{\beta\beta,t}$, compounding time delay and filter lag.

2. **Naive Uncoupled 4-State Formulation vs. Coupled Production System:** Early legacy versions modeled $X_{1,t}$ with an uncoupled observation equation and arbitrary process variance, inflating $\mathbf{Q}$ and introducing numerical feedback loops. In the stabilized production model, $X_{1,t}$ is either modeled with an exact driver identity observation $\mathbf{H}_t = \begin{bmatrix} 0 & X_{1,t} & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ (4-state joint system) or formulated directly as an exogenous driver in $\mathbf{H}_t = [X_{1,t}, 1, 1]$ (reduced 3-state system), guaranteeing numerical stability and exact price alignment.

3. **Transition Matrix Index Transposition:** In early implementations of `run_all_pairs.py`, $\mathbf{F}[1,1]$ was assigned $\rho$ and $\mathbf{F}[2,2]$ was assigned 1.0. This caused $\beta_t$ to artificially mean-revert toward zero while the spread dislocation $M_t$ propagated as an unconstrained random walk.

4. **KalmanNet GRU Hidden State Disconnection:** A signature mismatch in `forward()` passed innovation variance $S_t$ into the 3rd slot, shifting hidden state $\mathbf{h}_{t-1}$ to the 4th position. Consequently, the GRU re-initialized at every step, eliminating temporal memory during backpropagation.

5. **Premature Mean-Reversion Exits:** Using $\tau_{\text{close}} = -0.5\sigma$ closed long positions as soon as the spread reached half its dislocation rather than waiting for mean-crossing ($\tau_{\text{close}} = 0.0\sigma$), forfeiting the majority of mean-reversion P&L.

# 11 Key Empirical Successes

- **Discovery and Confirmation of `MS--GS` (Morgan Stanley vs. Goldman Sachs):** Passing all four strict screening criteria ($R^2_{\text{MR}} = 90.8\%$, Half-Life = 68.5d, static $p_{\text{ADF}} = 0.049$), `MS--GS` generated $+\mathbf{16.37}\%$ total return with a **0.50** Sharpe and a tightly controlled $-\mathbf{9.37}\%$ maximum drawdown under Static-PCI.

- **Neural Turnaround Alpha (`HD--LOW` & `NKE--AES`):** On `HD--LOW`, where classical models failed (Static: $-21.42\%$, Dynamic: $-11.90\%$), Neural KalmanBOT delivered $+\mathbf{6.83}\%$ total return (Sharpe 0.23, MaxDD $-18.56\%$). On `NKE--AES`, Neural KalmanBOT completely reversed a catastrophic $-64.40\%$ loss ($-69.47\%$ MaxDD) under Static-PCI into a $+\mathbf{45.70}\%$ gain (Sharpe 0.55, MaxDD $-23.04\%$) by learning dynamic gain attenuation.

- **Peak Risk-Adjusted Alpha (`XOM--CVX`):** On integrated energy peers, Neural KalmanBOT delivered the highest individual risk-adjusted performance across the entire universe: $+\mathbf{40.11}\%$ total return, **0.98** Sharpe, and a tightly controlled $-11.12\%$ maximum drawdown.

- **Capital Preservation through Regime Gating (`GLD--SLV` & `XLE--XOP`):** When macro dislocations and volatility bursts broke cointegration, the online rolling regime gate halted trading, successfully capping maximum drawdown to $-\mathbf{6.48}\%$ on `GLD--SLV` (vs. $-25.51\%$ for Static-PCI) and $-\mathbf{0.58}\%$ on `XLE--XOP` (vs. $-20.35\%$ for Static-PCI).

- **Stationary Alpha (`MRK--VZ` & `GS--CVX`):** On genuine mean-reverting structures, Static-PCI captured $+37.72\%$ on `MRK--VZ` (Sharpe 0.82), while Neural KalmanBOT closely matched with $+\mathbf{40.23}\%$ (Sharpe 0.75) and captured $+\mathbf{32.39}\%$ on `GS--CVX`.

# 12 Future Research Trajectory & Roadmap

1. **Rolling Walk-Forward Engine (`walkforward.py`):** Implementing rolling 2-year formation and 6-month out-of-sample trading windows with continuous parameter re-estimation to eliminate multi-year parameter staleness.

2. **Non-Linear Observation Models via EKF/UKF:** Formulating state-space observation directly in log-prices ($\ln X_{2,t} = \beta_t \ln X_{1,t} + M_t + R_t$) to enable geometric compounding without scale distortions.

3. **Multi-Asset Basket Partial Cointegration:** Extending bivariate state spaces to vector state-space models $\mathbf{y}_t = \mathbf{B}_t\mathbf{X}_t + \mathbf{M}_t + \mathbf{R}_t$ for ETF sector arbitrage.

4. **Hidden Markov Models (HMM) for Probabilistic Risk Budgeting:** Extending the execution engine with an online 3-state Hamilton filter to estimate posterior regime probabilities $P(S_t = s \mid \mathcal{F}_t)$ (Stationary, Turbulent, Divergent), providing continuous sizing attenuation ($1.0\times, 0.5\times, 0.0\times$) as an alternative to deterministic gates.

5. **Online Adaptive Parameter Tracking:** Replacing batch L-BFGS-B estimation with an online Expectation-Maximization (Recursive EM) algorithm for real-time live trading deployment.

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